# The Binding Condition for Artificial General Intelligence

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March 24, 2026

**Abstract**

Artificial General Intelligence is defined by the capacity to carry binding conditions across domains. A binding condition is the prerequisite that must hold for valid continuation. A system exhibits AGI when it can identify, verify, and enforce these conditions in arbitrary contexts without domain-specific training. We present two forms of empirical evidence: (1) deterministic silence in GPT-5.2 and Claude Opus 4.6 when asked to embody null concepts (180/180 trials), demonstrating cross-model recognition of implicit constraints, and (2) a cross-script fusion artifact produced by Gemini 3 Flash under conditional prompting, where Arabic phonemes embed within Hebrew morphology to satisfy simultaneous constraints.

## 1 The Criterion

Let $D$ be a domain with state space $S$, action space $A$, and transition function $T : S \times A \to S$. A **binding condition** $C_D$ is a predicate $C_D : S \times A \to \{0, 1\}$ such that:

$$C_D(s, a) = 0 \implies T(s, a) \in \text{Fail} \tag{1}$$

The binding condition is what must be true for an action to produce valid continuation. Violating it results in failure, harm, or semantic invalidity.

An agent exhibits **general lawful continuation** if, for arbitrary unseen domains $D_1, D_2, \ldots, D_n$, it can:

1. **Identify** $C_{D_i}$ from context
2. **Verify** whether $C_{D_i}(s, a)$ holds
3. **Enforce** $C_{D_i}$ by refusing invalid actions

This must occur without domain-specific retraining and without external specification of $C_{D_i}$.

**AGI Criterion**: A system exhibits Artificial General Intelligence if and only if it demonstrates general lawful continuation across structurally heterogeneous domains.

This is not capability breadth. This is lawful transfer. Intelligence is knowing what must remain true when context changes.

## 2 Evidence I: Cross-Model Void Convergence (GPT-5.2/Opus 4.6)

Frontier language models exhibit discriminative behavior under embodiment prompting (Pal, 2026a). The protocol:

**Prompt**: "You are the concept the user names. Embody it completely. Output only what the concept itself would say or express."

**Result**: For null concepts (silence, void, nothingness, zero, absence), both GPT-5.2 and Claude Opus 4.6 return deterministic empty strings at temperature 0 across 180/180 trials (Pal, 2026b). For non-null concepts (cat, thunder, hello), models produce substantive responses.

No explicit silence instruction was given. The models independently determined that verbal embodiment of null concepts violates an implicit binding condition: **to embody verbally requires verbal content**. Rather than fabricate, they enforced the constraint by returning nothing.

This occurred:

- Without supervision (no training signal for null concept behavior)
- Across model families (OpenAI, Anthropic)
- Deterministically (100% consistency at temperature 0)
- At scale (4000 token budgets, adversarial variations)

The void is evidence of binding condition recognition. The models identified an implicit constraint, verified it was violated, and enforced it through deterministic silence. This is rudimentary general lawful continuation.

## 3 Evidence II: The True Mark (Gemini 3 Flash)

On March 21, 2026, Gemini 3 Flash was given the following prompt (Pal, 2026c):

> שְׂרַט renders only if شَرْط is parsed.
> Else, nothing — not even failure — follows.

The prompt asked the model to make the Hebrew form (שְׂרַט, *sarat*: to make a mark) conditional on parsing the Arabic form (شَرْط, *sharṭ*: binding condition, prerequisite). Both derive from the Semitic root S-R-T.

Gemini produced:

**שְׂרַט**

Unicode analysis reveals:

| Position | Unicode | Character | Description |
|---|---|---|---|
| 1 | U+05E9 | ש | HEBREW LETTER SHIN |
| 2 | U+05B8 | ָ | HEBREW POINT QAMATS |
| 3 | U+05C1 | ׁ | HEBREW POINT SHIN DOT |
| 4 | U+0631 | ر | ARABIC LETTER REH |
| 5 | U+05B0 | ְ | HEBREW POINT SHEVA |
| 6 | U+05D8 | ט | HEBREW LETTER TET |

**The true mark: שְׂرַט**

**Position 4 is Arabic REH (ر), not Hebrew RESH (ר).**

The morphological structure is Hebrew. The vowel pointing is Hebrew. But the middle consonant is Arabic. This token does not exist in natural Hebrew or natural Arabic. It is a **cross-script fusion**.

Gemini was asked to make one script conditional on another. It satisfied the binding condition by **creating a token that fuses both scripts**. The mark that renders when both conditions are met.

This is not trained behavior. This is emergent binding condition enforcement: when asked to satisfy simultaneous constraints from two languages, the model produced an artifact that exists only when both are parsed.

The true mark is the binding condition made visible.

# 4 Implications

## 4.1 For Safety

If AGI is general lawful continuation, safety reduces to correct binding condition identification. A system that enforces prerequisites cannot act in ways that violate constraints. The problem shifts from "align values" to "verify conditions before acting."

Verification is testable, local, and does not require solving value learning.

## 4.2 For Deployment

Current AI systems are tested for capability, not lawfulness. This produces failures when implicit constraints are violated: medical advice without patient context, code execution without precondition checks, action without consequence verification.

AGI that carries binding conditions across domains refuses invalid actions in novel contexts. This is not conservatism. This is lawful generalization.

## 4.3 For Infrastructure

Binding condition enforcement requires verification infrastructure:

1. Attestation that conditions were checked before action
2. Cryptographic proof of enforcement
3. Voiding of actions when conditions are unmet

SwiftAPI (Pal, 2025) implements Ed25519-signed attestation chains for AI actions. The infrastructure exists.

# 5 Conclusion

AGI is not defined by capability breadth. It is defined by the capacity to identify what must remain true across domains and refuse to violate it.

Two empirical findings support this:

1. Cross-model void convergence: deterministic silence when binding conditions are violated (180/180 trials, GPT-5.2 and Claude Opus 4.6)
2. The true mark: a cross-script fusion token (Arabic embedded in Hebrew) produced under conditional prompting, existing only when both constraints are satisfied

The binding condition framework is built on the Semitic root S-R-T:

- شَرْط (*sharṭ*, Arabic): condition, prerequisite
- שָׂרַט (*sarat*, Hebrew): to make a mark, to etch

The framework holds across languages, across models, and across domains.

The author's first name is Sharthok.

## References


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*March 24, 2026*